Hold Onto That Thought: Assessing KV Cache Compression On Reasoning

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NEURAL INFORMATION PROCESSING SYSTEMS

TLDR: KV compression techniques behave *very differently* on long reasoning traces than on long prompts.

Heavy-hitter attention methods (SnapKV-D, H2O) are the only ones that consistently preserve accuracy, while many popular approaches slow models down or break reasoning entirely.

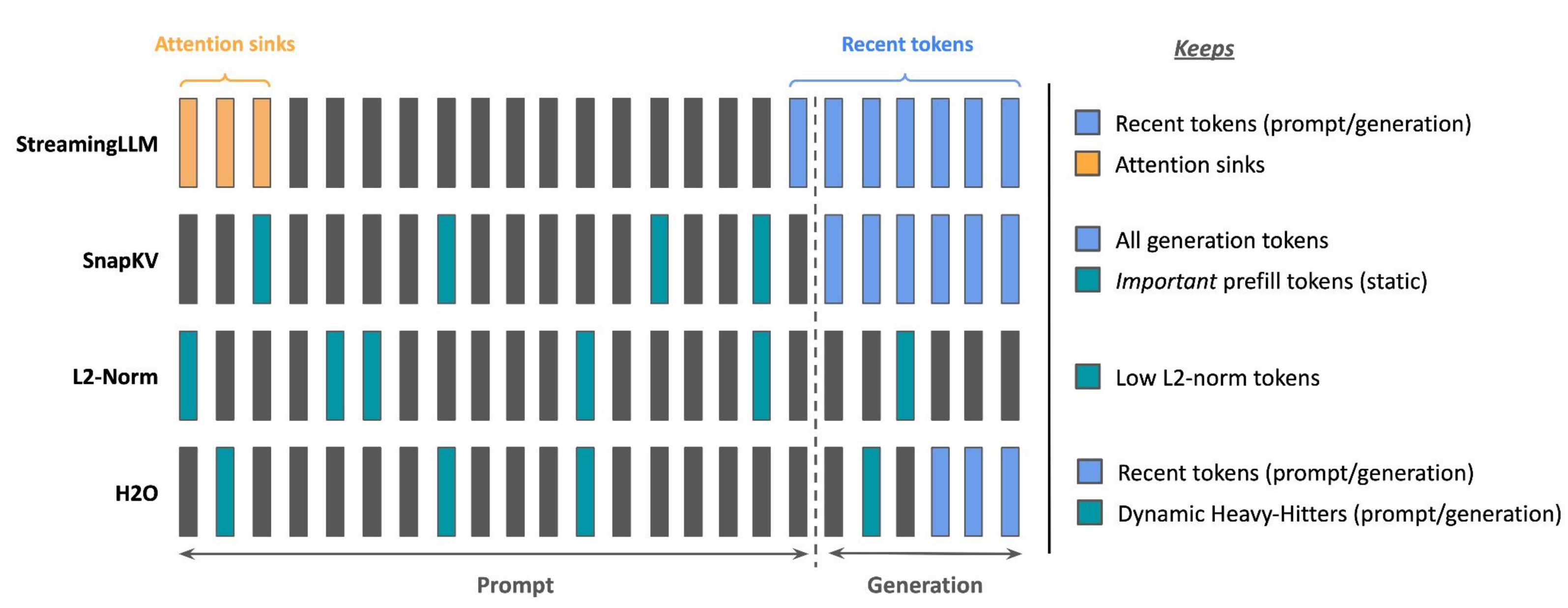
- **Reasoning sequences—not prompts—dominate KV cache usage** in modern distilled R1-style models.
- **SnapKV-D and H2O outperform all other compression methods** across tasks and budgets.
- **Some methods make models "think longer"**—compressed caches can *increase* output length and inference cost.
- **No universal winner** for non-reasoning models (e.g., Llama-3.1-8B-Instruct).
- **Attention-based token importance is the most reliable signal** for safe KV compression in reasoning workloads.

Motivation

Why KV Compression?

- LLM inference stores all previous Key/Value vectors to avoid recomputation.
- KV cache grows linearly → memory bottleneck on single-GPU systems.
- Most cache compression work targets long prompts, not long decoding.

Reasoning models (DeepSeek-R1, Nemotron, etc.) generate thousands of tokens in thinking traces, even when prompts are short.
These long decoding sequences stress KV memory far more than long prompts.



A Conceptual Comparison of Token Retention Strategies in Different KV Cache Compression Methods. Each row illustrates a method's logic for retaining tokens (colored) versus evicting them (gray) from the KV cache during a long sequence divided into a prefill and decoding phase.

Main Findings

Heavy-Hitter Methods Win for Reasoning

Across reasoning models (DeepSeek-R1, Nemotron):

SnapKV-D and H2O are the dominant strategies across all budgets and datasets.

Reasons:

- Both track **accumulated attention**, which strongly correlates with importance.
- SnapKV-D's sliding window captures **emergent heavy hitters during reasoning**, not only from the prompt.

Non-Reasoning Model Shows No Universal Winner

Llama-3.1-8B-Instruct:

- Sometimes StreamingLLM is best (math). Sometimes SnapKV-D / H2O take over.
- *Reason:* Non-reasoning outputs have **lower density of critical tokens**, so heuristics vary.

Latency & Efficiency

- StreamingLLM and KNorm are fastest → no attention accumulation.
- SnapKV-D slower at very small budgets (frequent evictions).
- H2O incurs mild overhead.
- **Throughput differences are minor compared to accuracy differences.**

Output Length Explosion

- Compression can cause **longer reasoning traces**, especially at low budgets.
- KNorm often causes **non-terminating looped thinking**.
- Explains why accuracy sometimes *decreases* even when tokens saved.

Experiment Results

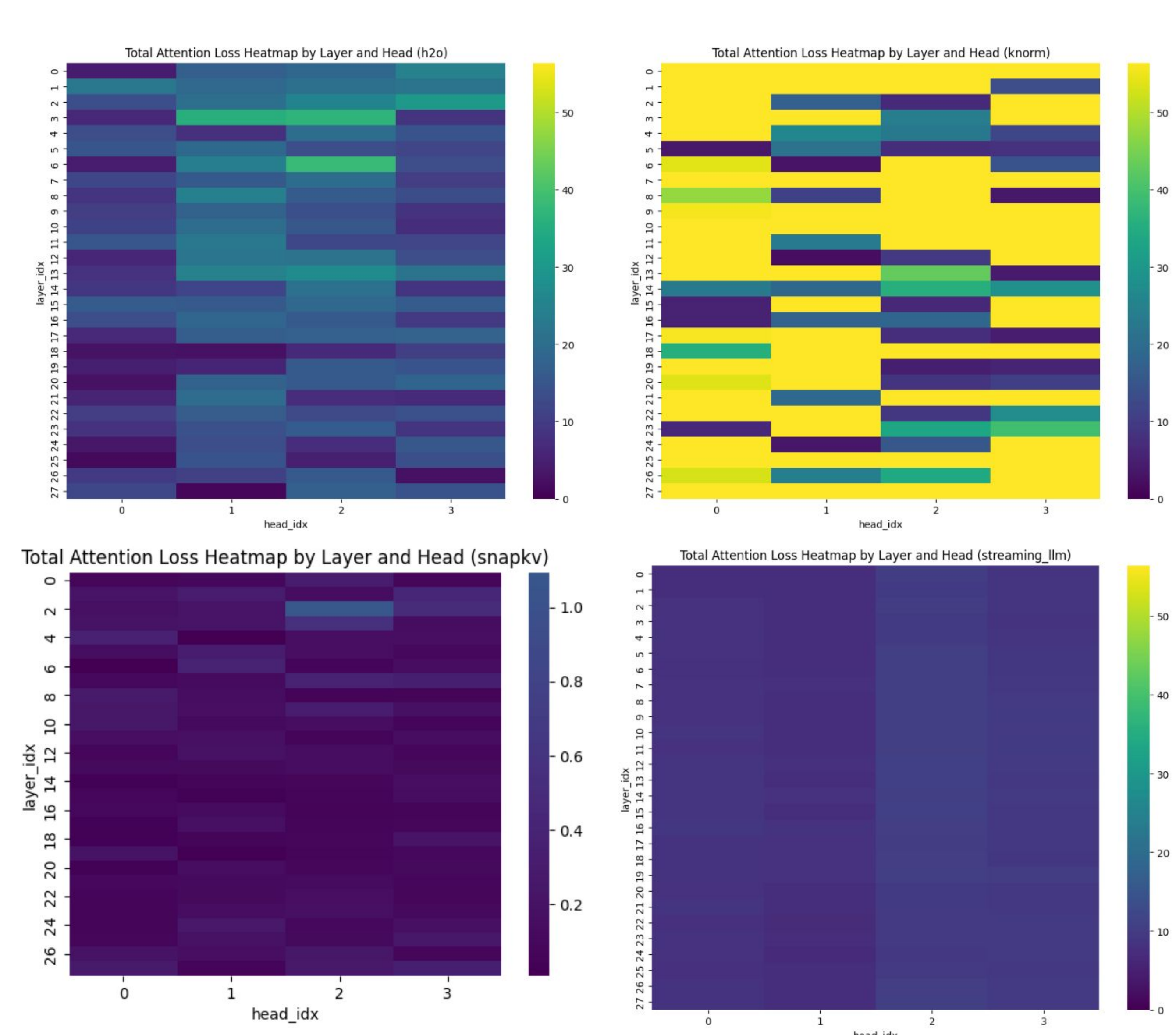
Llama-3.1-8B-Instruct	GSM8K				Math500				CSQA				OBQA			
	128	256	384	512	128	256	384	512	128	256	384	512	128	256	384	512
Full	0.88				0.39				0.77				0.84			
ShadowKV	0.32				0.22				0.20				0.31			
H2O	**0.63**	**0.77**	0.82	0.83	**0.30**	**0.33**	**0.33**	**0.36**	**0.74**	0.76	**0.77**	**0.77**	**0.83**	**0.86**	**0.86**	**0.86**
Knorm	0.05	0.53	0.73	0.82	0.03	0.18	0.22	0.33	0.34	**0.77**	0.75	0.76	0.41	0.79	0.84	0.82
RKV	0.12	0.34	0.50	0.49	0.03	0.10	0.16	0.20	0.36	0.62	0.76	0.77	0.22	0.66	0.77	0.84
SnapKV	0.53	0.55	0.56	0.53	0.20	0.21	0.19	0.20	0.70	0.64	0.71	0.72	0.73	0.77	0.72	0.76
StreamingLLM	0.26	0.75	**0.84**	**0.87**	0.11	0.26	0.32	0.35	0.20	0.75	0.76	**0.77**	0.14	0.72	0.84	0.84
	ReClor				DROP				StrategyQA				FOLIO			
	128	256	384	512	128	256	384	512	128	256	384	512	128	256	384	512
Full	0.60				0.15				0.83				0.51			
ShadowKV	0.27				0.28				0.68				0.33			
H2O	0.32	0.56	**0.60**	0.58	0.12	**0.14**	**0.17**	**0.17**	**0.81**	**0.87**	0.88	**0.89**	0.22	**0.43**	0.41	0.43
Knorm	0.01	0.19	0.46	**0.59**	0.01	0.08	0.13	0.13	0.47	0.85	0.88	0.87	0.02	0.28	0.39	0.38
RKV	0.04	0.21	0.40	0.54	0.06	0.07	0.14	0.11	0.60	0.79	0.77	0.79	0.07	0.36	0.44	0.34
SnapKV	**0.53**	**0.57**	0.58	0.55	**0.15**	0.12	0.11	0.12	0.78	0.78	0.81	0.76	**0.44**	0.40	**0.45**	**0.46**
StreamingLLM	0.05	0.21	0.59	0.58	0.09	0.11	0.15	0.16	0.11	0.76	**0.89**	0.85	0.03	0.09	0.25	0.35

Deepseek-R1-Distill-Qwen-7B	GSM8K				Math500				CSQA				OBQA			
	128	256	384	512	128	256	384	512	128	256	384	512	128	256	384	512
Full	0.70				0.47				0.67				0.78			
ShadowKV	0.47				0.33				0.20				0.31			
H2O	0.21	0.44	0.51	0.52	0.14	0.21	0.29	0.31	0.44	0.61	**0.60**	**0.64**	0.42	0.64	**0.69**	0.67
Knorm	0.00	0.00	0.08	0.16	0.00	0.01	0.03	0.05	0.05	0.13	0.30	0.42	0.03	0.05	0.23	0.38
RKV	0.04	0.07	0.18	0.30	0.04	0.04	0.05	0.17	0.10	0.09	0.27	0.34	0.10	0.10	0.21	0.26
SnapKV	**0.67**	**0.67**	**0.70**	**0.71**	**0.38**	**0.36**	**0.36**	**0.32**	**0.65**	**0.62**	0.59	0.61	**0.71**	**0.75**	0.68	**0.76**
StreamingLLM	0.02	0.19	0.32	0.44	0.03	0.12	0.19	0.26	0.08	0.14	0.31	0.48	0.02	0.11	0.28	0.37
	ReClor				DROP				StrategyQA				FOLIO			
	128	256	384	512	128	256	384	512	128	256	384	512	128	256	384	512
Full	0.45				0.16				0.67				0.36			
ShadowKV	0.27				0.14				0.60				0.33			
H2O	0.01	0.04	0.18	0.28	0.04	0.07	0.10	0.10	0.33	**0.64**	**0.74**	**0.72**	0.03	0.21	0.23	0.23
Knorm	0.00	0.00	0.01	0.01	0.00	0.01	0.01	0.03	0.00	0.12	0.44	0.59	0.00	0.01	0.03	0.05
RKV	0.04	0.03	0.02	0.01	0.04	0.04	0.03	0.04	0.05	0.14	0.34	0.42	0.04	0.03	0.02	0.06
SnapKV	**0.45**	**0.39**	**0.40**	**0.43**	**0.13**	**0.11**	**0.12**	**0.16**	**0.60**	0.59	0.57	0.63	**0.30**	**0.25**	**0.31**	**0.29**
StreamingLLM	0.00	0.01	0.01	0.04	0.04	0.05	0.08	0.13	0.00	0.05	0.22	0.42	0.00	0.01	0.02	0.03

Nemotron-Nano-8B	GSM8K				Math500				CSQA				OBQA			
	128	256	384	512	128	256	384	512	128	256	384	512	128	256	384	512
Full	0.64				0.45				0.51				0.64			
ShadowKV	0.44				0.28				0.20				0.31			
H2O	0.22	0.45	0.52	0.57	0.16	0.24	0.31	0.33	0.47	0.49	**0.52**	0.51	0.59	0.59	0.58	0.62
Knorm	0.01	0.02	0.09	0.18	0.01	0.01	0.03	0.06	0.36	0.40	0.44	0.46	0.32	0.44	0.48	0.57
RKV	0.04	0.03	0.09	0.15	0.02	0.04	0.03	0.06	0.28	0.30	0.42	0.41	0.35	0.44	0.51	0.51
SnapKV	**0.65**	**0.63**	**0.66**	**0.66**	**0.41**	**0.44**	**0.45**	**0.43**	**0.49**	**0.50**	0.51	**0.53**	**0.68**	**0.63**	**0.66**	**0.66**
StreamingLLM	0.03	0.20	0.40	0.53	0.02	0.13	0.22	0.34	0.36	0.44	0.46	0.50	0.36	0.46	0.52	0.62
	ReClor				DROP				StrategyQA				FOLIO			
	128	256	384	512	128	256	384	512	128	256	384	512	128	256	384	512
Full	0.48				0.11				0.89				0.36			
ShadowKV	0.27				0.11				0.65				0.33			
H2O	0.20	0.22	0.35	**0.40**	0.05	0.06	0.10	0.09	0.76	0.84	**0.85**	**0.83**	0.22	0.36	0.35	0.37
Knorm	0.01	0.03	0.07	0.07	0.01	0.01	0.02	0.03	0.38	0.55	0.68	0.76	0.03	0.04	0.07	0.13
RKV	0.03	0.08	0.08	0.07	0.02	0.06	0.05	0.03	0.42	0.45	0.64	0.71	0.06	0.08	0.11	0.14
SnapKV	**0.42**	**0.42**	**0.42**	0.37	**0.11**	**0.11**	**0.12**	**0.10**	**0.83**	**0.85**	0.84	**0.84**	**0.38**	**0.42**	**0.41**	**0.41**
StreamingLLM	0.03	0.06	0.09	0.14	0.03	0.02	0.06	0.08	0.24	0.39	0.52	0.69	0.03	0.03	0.06	0.15

DeepSeek-R1-Distill-Llama-8B	GSM8K				Math500				CSQA				OBQA			
	128	256	384	512	128	256	384	512	128	256	384	512	128	256	384	512
Full	0.70				0.46				0.75				0.84			
ShadowKV	0.51				0.34				0.20				0.31			
H2O	0.37	0.53	0.62	0.61	0.20	0.31	0.36	0.36	0.48	0.72	0.73	**0.73**	0.48	0.78	**0.83**	**0.84**
Knorm	0.00	0.09	0.19	0.28	0.00	0.01	0.02	0.06	0.05	0.28	0.54	0.66	0.03	0.27	0.57	0.70
RKV	0.05	0.04	0.14	0.17	0.03	0.05	0.02	0.02	0.07	0.11	0.16	0.35	0.07	0.07	0.19	0.32
SnapKV	**0.72**	**0.72**	**0.74**	**0.72**	**0.42**	**0.44**	**0.41**	**0.41**	**0.74**	**0.73**	**0.74**	**0.73**	**0.82**	**0.83**	**0.83**	0.81
StreamingLLM	0.06	0.25	0.39	0.56	0.03	0.09	0.21	0.29	0.04	0.14	0.35	0.50	0.07	0.15	0.32	0.52
	ReClor				DROP				StrategyQA				FOLIO			
	128	256	384	512	128	256	384	512	128	256	384	512	128	256	384	512
Full	0.51				0.14				0.74				0.47			
ShadowKV	0.27				0.09				0.80				0.33			
H2O	0.03	0.08	0.23	0.38	0.06	0.07	0.10	0.11	0.25	**0.69**	**0.77**	**0.79**	0.07	0.37	0.41	**0.46**
Knorm	0.00	0.00	0.02	0.10	0.00	0.01	0.01	0.05	0.06	0.36	0.57	0.70	0.00	0.03	0.11	0.21
RKV	0.04	0.03	0.03	0.11	0.03	0.03	0.05	0.07	0.08	0.35	0.50	0.63	0.03	0.08	0.13	0.26
SnapKV	**0.52**	**0.53**	**0.56**	**0.51**	**0.17**	**0.15**	**0.15**	**0.16**	**0.68**	0.66	0.64	0.68	**0.46**	**0.45**	**0.49**	**0.46**
StreamingLLM	0.00	0.00	0.01	0.06	0.02	0.02	0.09	0.13	0.03	0.11	0.36	0.56	0.00	0.01	0.04	0.09

Accuracy of all KV-compression methods across four budget settings (128–512). Results are averaged over benchmark tasks. Heavy-hitter attention methods (SnapKV-D, H2O) consistently yield the highest accuracy under tight budgets, while heuristic and recency-based approaches degrade substantially as compression increases.



Attention Loss Heatmaps. Lower attention loss → higher accuracy. SnapKV-D retains the most important tokens.